DEFENCE · SECURITY & RESILIENCE

Know if the Wi-Fi is safe before you trust it.

NetGuard checks the security posture of every network you join, then tells you in plain language whether it can be trusted.

THE PROBLEM

Public Wi-Fi asks for trust it never earned.

The network that looks ordinary can be the one reading your traffic.

1 click

to join a network that is not what it claims to be.

0 signals

that your traffic is being read or rewritten.

Everyone

remote workers, small teams, travellers. No security team required.

NetGuard acts at the exact moment of risk: the network change.

WHY THE USUAL ANSWERS FAIL

Existing help arrives too late, or says too little

ENTERPRISE TOOLS	**Built for managed fleets,** too heavy for one laptop on hotel Wi-Fi.
STANDARD ADVICE	**"Just use a VPN",** which never tells you whether this network is hostile.
THE OPERATING SYSTEM	**A padlock icon,** which says nothing about who else is on the network.

The insight: make the network change the trigger, and turn ten checks into one verdict.

IDEA & INNOVATION

Check the network at the moment you join it

01 · DETECT

Watch the network

The moment you connect, **NetGuard notices**, and starts looking.

02 · ASSESS

Run ten checks

Is the link protected? Who runs it? **Is anyone in the middle?**

03 · INFORM

Give one verdict

Safe, caution or unsafe, **with one clear action**, right on the desktop.

Proactive by trigger, explainable by design, reproducible without special hardware.

HOW IT WORKS

A small, transparent pipeline

You connect
a new network appears

→

Facts are gathered
what kind of network, and who runs it

→

Ten checks run
protection, identity, eavesdropping

→

Findings are weighed
one score out of 100

→

You are told
a verdict and one clear action

Every check inspects **the network, not your files**: how the connection is protected, whether the router is who it claims to be, and whether anyone is quietly reading your traffic. **Nothing is captured, recorded or sent anywhere.**

THE POSTURE CHECKS

Ten questions, asked every time you connect

Wireless encryption — HIGH
The link is open, or protected by long-broken security.

Intercepted traffic — CRITICAL
Someone is standing between you and the sites you visit.

Faked name lookups — HIGH
The network lies about where websites really live.

Impersonated gateway — HIGH
The router's identity does not match what the network claims.

Twin hotspots — HIGH
Another access point borrows the same network name.

Forced login pages — MEDIUM
Traffic is held until you sign in on the network's page.

Hidden relays — MEDIUM
An unseen middleman passes unencrypted traffic through.

Open management — MEDIUM
The router's admin doors are reachable by anyone nearby.

Traceable device — LOW
Your device broadcasts a permanent, trackable identifier.

Working connection — INFO
Confirms the internet is actually reachable.

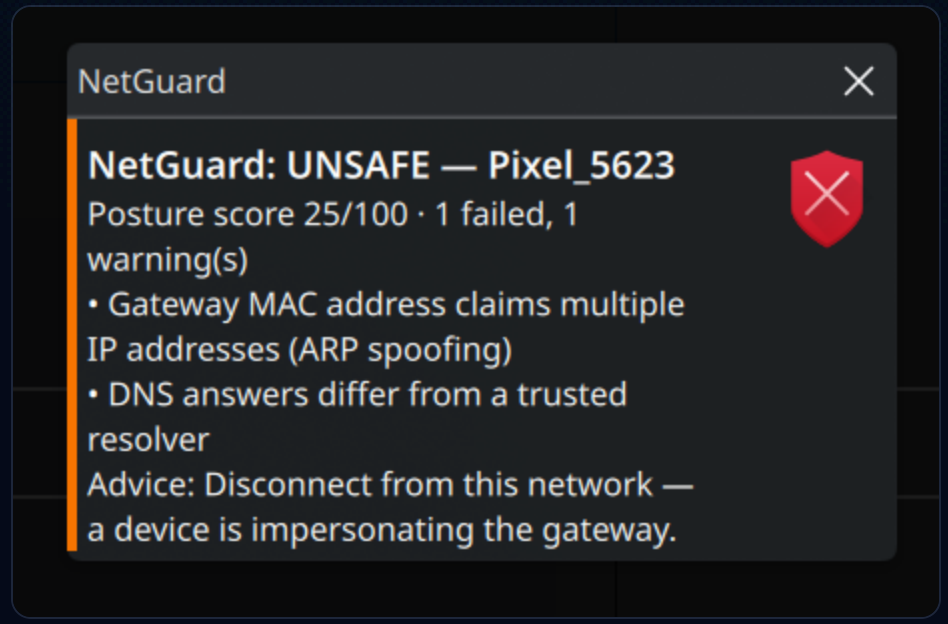
When a check cannot run, it is **skipped rather than guessed**, so a blocked network never produces a false alarm.

VERDICT & SCORING

From ten answers to one decision



- **Serious means serious.** Every failure subtracts from 100 by how dangerous it is; warnings cost less.
- **No averaging away.** One critical finding holds the verdict at unsafe.
- **Reasons and action.** A plain verdict, the problems behind it, and one recommended action.



A real verdict on a hostile network.

WHAT YOU ACTUALLY SEE

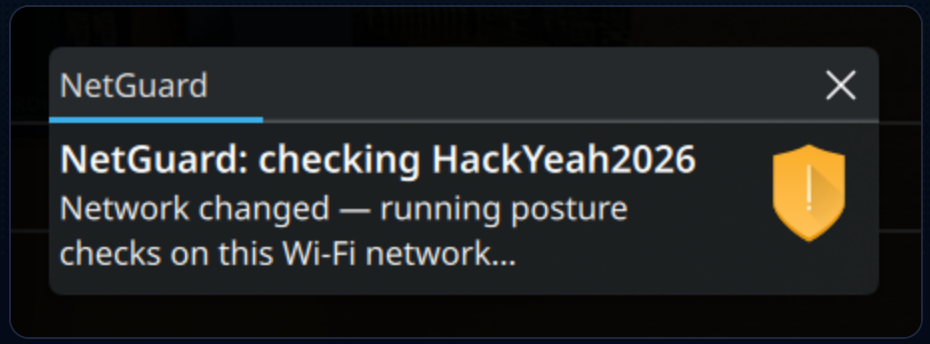
Three seconds after joining, you know

1 · YOU CONNECT

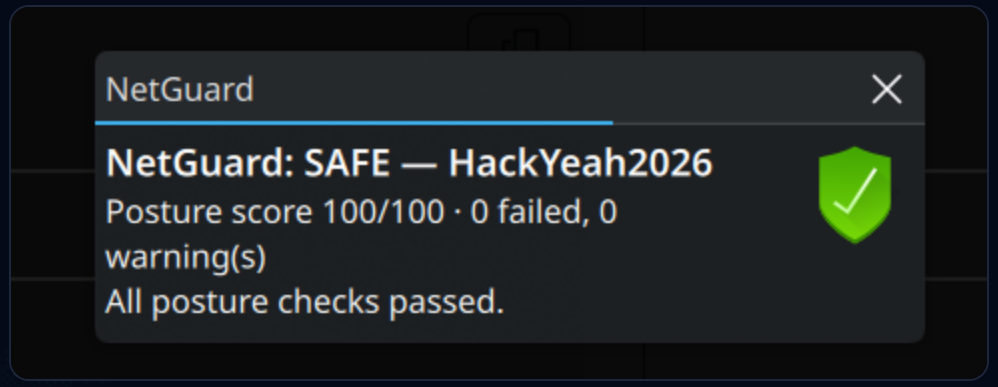
HackYeah2026

Wi-Fi connected · network changed

2 · NETGUARD CHECKS



3 · YOU GET THE VERDICT



Safe networks get a quiet green confirmation. **An unsafe verdict stays on screen until you dismiss it**, and a full report is written for anyone who wants the detail.

DEMONSTRATION

Six networks, six verdicts

SCENARIO	THE SITUATION	WHAT NETGUARD FINDS	VERDICT
home-safe	A well-protected home network	Everything passes	SAFE
wired-office	An office cable connection	Wireless checks not applicable	SAFE
cafe-open	An open café hotspot	No encryption, forced login page	CAUTION
airport-captive	Airport Wi-Fi	No encryption, forced login, open router admin	CAUTION
corp-interception	A managed company network	The company can read encrypted traffic	CAUTION
hotel-mitm	A compromised hotel network	Fake hotspot, faked lookups, intercepted traffic	UNSAFE

The simulator swaps only the data: **the same checks, scoring and notifications** run exactly as they do live, so the whole flow can be shown on any machine.

PRACTICAL APPLICABILITY

Built to deploy on a typical machine

- **Nothing to install.** Uses what your operating system already has, and nothing else.
- **Runs quietly in the background.** Starts with your session, no administrator rights.
- **Only speaks up when it matters.** It works when the network changes, and skips what it cannot see rather than guessing.

No capture

Nothing about your traffic is recorded or stored.

No cloud

Nothing leaves the machine, and every line is auditable.

HONESTY & ROADMAP

What it does well, and what comes next

KNOWN LIMITATIONS

- Spotting a **cloned hotspot** is the hardest part; it needs deeper wireless fingerprinting.
- A company that installed its own security certificate on your laptop is reported as a warning.

ROADMAP

- Deeper wireless fingerprinting to **catch cloned hotspots**.
- Stronger protection for name lookups, and per-app insight.
- A small menu-bar app, and a fleet mode for teams.

DEFENCE · SECURITY & RESILIENCE

Make the safe choice automatic.

NetGuard turns the instant of a network change into an automatic security posture check, so people, not just enterprises, can tell when public Wi-Fi is not safe to use.